

HOMEWORK 3

1. Prove that a Dedekind ring with finitely many maximal ideals is a PID.
2. Let R be a Dedekind domain and let I and J be nonzero ideals of R . Show that I and J are coprime (i.e., $I + J = R$) if and only if they do not have common prime ideal in their factorizations into products of prime ideals.
3. Let P and Q be distinct prime ideals of a Dedekind domain. Show that $P \cdot Q = P \cap Q$.
4. Prove that every ideal of a Dedekind ring is generated by at most two elements.
5. Prove that every UFD is a normal ring.
6. Let R be a Dedekind ring and $a \in R$ a nonzero element. Prove that the ring R/aR is Artinian.
7. Let I and J be nonzero ideals of a Dedekind ring R . Prove that the R -modules I/IJ and R/J are isomorphic.
8. Let R be a ring of algebraic integers in a finite field extension of $\mathbb{Q}$ and $p \in \mathbb{Z}$ a prime integer. Prove that the set A of all prime ideals $P \subset R$ such that $P \cap \mathbb{Z} = p\mathbb{Z}$ is finite and nonempty. Show that $pR = \prod P^{e_P}$ for some positive integers e_P , where the product is taken over all $P \in A$.
9. In the conditions of the previous problem let n be the degree of the quotient field of R over $\mathbb{Q}$. Show that for every $P \in A$, the quotient ring R/P is a finite field of p^{f_P} elements for some integer f_P . Prove that $n = \sum e_P f_P$, where the sum is taken over all $P \in A$. (Hint: Compute the number of elements in R/pR .)
10. Let $\alpha = (1 + \sqrt{-23})/2$ and let I be the ideal in $\mathbb{Z}[\alpha]$ generated by 2 and α . Prove that I^3 is a principal ideal.